

Joe's Snacks Network Design

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Overview

In pursuit of my Master's degree in Cybersecurity, I have familiarized myself with the threats that successful snack companies are subject to. Threat actors disguised as customers seeking to steal private customer information. Unsuspecting employees unfamiliar with the dangers of the internet and how malware is obtained and spread. I have been hard at work to design a network that meets your needs. My manufacturing background in the biotech field has granted me hands-on experience with sophisticated network topologies in an enterprise setting. I have acquired experience as an end user for technology commonly implemented by growing companies such as yours. Mr. Joseph Jasper, in this report I aim to provide you with a network design that is secure, scalable, and efficient.

This Joe's Snack network has been designed to meet the needs of your growing company, customers, and employees. The network architecture will include Dual-Path Dual-Layer firewall network architecture. A "Red Rail" path will be utilized to allow new inbound TCP connections while the "Green Rail" path acts as a pathway that exclusively allows new TCP connections in the outgoing direction. The IP addressing includes five unique subnets chosen in consideration of the current size of your company and its potential for scalability. DMZ-Subnet, Intranet-Subnet, Internal-Subnet, R1-Subnet and the G1-Subnet have been appropriately sized to provide private IP addresses for your devices. Four servers and two gateways have been placed exclusively in either the DMZ-Subnet and Intranet-Subnets with considerations to both reachability and security. Network Address Translation (NAT) has been configured on two public internet facing routers to support remote employee access, customer orders, and internal workstation public internet server access. Default static routes and non default static routes have been configured on all firewalls to support proper packet routing and device to device communication.

There were several security challenges that became prevalent when constructing this design. Public facing services such as JS-Sales-Server, the device which provides the means for e-commerce, needs to be accessible by virtually anyone on the internet. Having a publicly accessible server introduces risk and serves as a potential attack vector. Potential threats include web application attacks, Vulnerability exploitation, and Denial of Services attacks. The same case is made for the RD-Gateway which provides remote employees access to internal workstations and other services. With the RD-Gateway being internet accessible, the potential threat of credential attacks becomes especially relevant. Remote access systems are frequently targeted because they provide a direct path into corporate resources. As you continue reading, you will see just how these threats were mitigated. Other security concerns such as limited public address space, Internet access control, and protection of sensitive data will also be tackled in the upcoming sections. The methods of mitigating these security concerns

built into the design of this network provided in this report will also serve as evidence as to why I am in fact the most promising candidate to design the Joe's Snack network.

Network Architecture

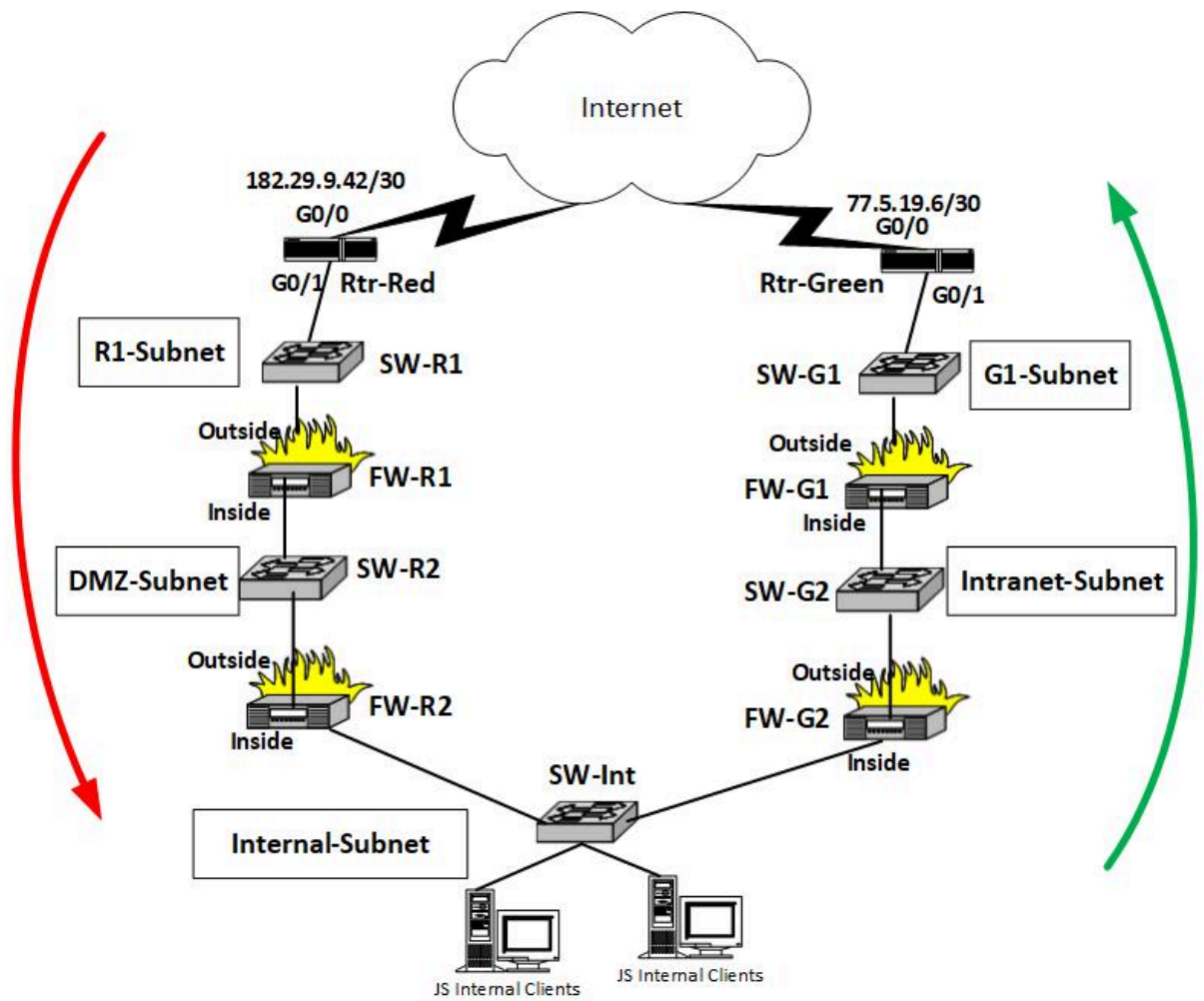
The implementation of the Dual-Path Dual-Layer network architecture offers access to public facing servers such as the servers used by employees working remotely and the server customers access to place orders. This is accomplished all while segregating the network so access to internal devices, like one you would use to keep important HR related files, is safe behind a firewall. The Dual-Path Dual-Layer architecture incorporates four firewalls. These firewall devices are strategically positioned on the network creating security zones that enforce segmentation restricting traffic to approved application flows. Switch stack sizes were selected by dividing the required number of device connections by 24 ports per Switch Unit and rounding up to the next whole Switch Unit. This ensures each subnet has enough physical ports to support the required devices and firewall/router links. SW-R1 only connects the Red router and FW-R1, so 1 SU is enough. SW-R2 located in the DMZ must support up to 40 wired servers, plus firewall links, so 2 SUs gives 48 ports. SW-Int needs 200 workstations + 20 Access Points + firewall links = 222 ports, so $222 \div 24 = 9.25$, rounded up to 10 SUs. SW-G2 located in the Intranet must support up to 40 wired servers plus firewall links, so 2 SUs gives 48 ports. SW-G1 only connects FW-G1 and Rtr-Green, so 1 SU is enough. This results in 16 total switch units. This particular network architecture can be viewed in terms of having a red rail, a green rail, and is much more suited for the company's needs than alternative architectures.

On the red rail sits a public internet facing router, two firewalls, and devices that are accessible by internet users. Each firewall contains 2 interfaces for which traffic can be appropriately routed or denied. The placement of the firewalls create security zones that help control what company assets are accessible. Flow of traffic on the red rail is restricted allowing new inbound TCP connections exclusively. When the red router receives a new TCP connection request from a source outside the network, it forwards that request to the first firewall. This potentially dangerous packet is examined before determining whether or not it can be allowed further into the network. It is beyond the first firewall where resources designed for public access are kept. The red rail is home to devices that provide services to public internet users. Establishing a connection to access these services requires permission from a firewall facing the internet. A second firewall on the red rail is configured to block unauthorized traffic from leaving the security zone and accessing a different zone. The devices that host the services provided to public internet users sit between these two firewalls. The law of least privilege is the guiding principle for which the firewalls examine every packet it receives. The green rail is home to more sensitive devices such as the servers containing private company data and resources. The firewalls that sit on the green rail, much like that of the red rail, enclose these devices on both sides permitting authorized access only. Only new outgoing TCP requests flow through the internet facing router on the green rail, unlike the router on the red rail that exclusively receives incoming TCP requests. Internal clients and workstations are placed in the middle of the red and green rails with the green rail providing access to public web servers. Separate red and green Internet paths were implemented to improve network resilience and segregate inbound and outbound traffic flows. Public-facing services are hosted on the red rail, while employee Internet access is routed through the green rail via the WSA-Gateway. This separation

reduces the impact of a single point of failure and simplifies security policy enforcement making it superior to other architectures such as two leg or three leg firewall architecture.

Two-Leg firewall architecture may be appealing in its simple design and low cost, however it fails to provide a necessary DMZ security zone for Joe's Snacks public facing servers. The absence of a DMZ security zone would require public facing systems to reside on the same network as systems intended for internal use only, drastically increasing the risk of an attacker compromising a web server and gaining direct access to internal resources. Two-Leg architecture also lacks redundancy and leaves the network vulnerable to single point failure, a disadvantage also observed in a competing architecture. While Three-leg architecture does implement a DMZ security zone, it leaves all security policies to be enforced by a single firewall increasing the rule complexity and likelihood of configuration errors. A single-path Dual-layer architecture requires all traffic, including public services and employee internet access to use a single internet path. This subjects the network to congestion, reduces resilience, and creates a larger attack surface. A single point of internet failure will impact all services. The Dual-Path Dual-Layer Firewall architecture implemented for the JS network reduces policy complexity, provides better traffic isolation, easier troubleshooting, and improved security monitoring. It provides the best balance of security, segmentation, availability, and traffic isolation for the JS network. Compared to two-leg, three-leg, and single-path dual-layer alternatives, the selected design better supports the organization's public-facing services, remote access requirements, employee Internet access, and defense-in-depth security strategy while minimizing the exposure of critical business systems.

JS Network Switches		
Switch Stack Name	Number of SUs	Comments
SW-R1	1	
SW-R2	2	
SW-Int	10	
SW-G2	2	
SW-G1	1	



The JS Network

IP Addressing

IP addressing was designed to avoid unnecessary IP waste. Private ip addresses falling under the 10.53.0.0/16 address block are implemented. Each subnet is sized based on the number of required devices. The JS network contains five subnets. The Internal-Subnet contains subnet ID 10.53.1.0/24 with subnet mask of 255.255.255.0. It has a capacity for 254 hosts with usable IP range of 10.53.1.1 to 10.53.1.254. The DMZ-Subnet was given subnet ID 10.53.2.0/26 with a subnet mask of 255.255.255.192. It has a capacity of 62 hosts with usable IP range of 10.53.2.1 to 10.53.2.62. The Intranet-Subnet has a subnet ID of 10.53.2.64/26 with a subnet mask of 255.255.255.192. It has the capacity to maintain 62 hosts with an IP range of 10.53.2.65 to 10.53.2.126. The R1-subnet possesses the subnet ID 10.53.2.128/29 with subnet mask 255.255.255.248. This smaller subnet has the capacity for 6 hosts and a usable IP range of 10.53.2.129 to 10.53.2.134. The G1-Subnet has the subnet ID 10.53.2.136/29 with the subnet mask 255.255.255.248. It has a capacity for 6 hosts and a usable IP range of 10.53.2.137 to 10.53.2.142.

The Internal-Subnet was assigned a /24 mask because it must support up to 200 wired workstations and 20 wired access points. A /24 provides 254 usable host addresses, which is sufficient for these devices and the required firewall and switch management interfaces. The DMZ-Subnet and Intranet-Subnet were each assigned /26 masks. Each /26 provides 62 usable host addresses, which is enough to support up to 40 wired servers plus firewall interfaces and switch management addresses. The R1-Subnet and G1-Subnet were assigned /29 masks because they are small transit networks connecting routers, firewalls, and switch management interfaces. A /29 provides 6 usable host addresses, which is enough for those infrastructure devices while conserving address space.

<u>JS Network IP Subnets</u>					
<u>Subnet Name</u>	<u>Network Address</u>	<u>Prefix Length (/n)</u>	<u>Number of Assignable IPs</u>	<u>First Assignable IP</u>	<u>Last Assignable IP</u>
<u>R1-Subnet</u>	10.53.2.128	/29	6	10.53.2.129	10.53.2.134
<u>DMZ-Subnet</u>	10.53.2.0	/26	62	10.53.2.1	10.53.2.62
<u>Internal-Subnet</u>	10.53.1.0	/24	254	10.53.1.1	10.53.1.254
<u>Intranet-Subnet</u>	10.53.2.64	/26	62	10.53.2.65	10.53.2.126
<u>G1-Subnet</u>	10.53.2.136	/29	6	10.53.2.137	10.53.2.142

JS Network Device Interface IPs				
Device Name	Interface	Subnet Name	IP Address	Subnet Mask
Rtr-Red	G0/0	ISP-Link1	182.29.9.42	255.255.255.252
Rtr-Red	G0/1	R1-Subnet	10.53.2.129	255.255.255.248
Rtr-Green	G0/0	ISP-Link2	77.5.19.6	255.255.255.252
Rtr-Green	G0/1	G1-Subnet	10.53.2.137	255.255.255.248
FW-R1	Inside	DMZ-Subnet	10.53.2.1	255.255.255.192
FW-R1	Outside	R1-Subnet	10.53.2.130	255.255.255.248
FW-R2	Inside	Internal-Subnet	10.53.1.1	255.255.255.0
FW-R2	Outside	DMZ-Subnet	10.53.2.2	255.255.255.192
FW-G2	Inside	Internal-Subnet	10.53.1.2	255.255.255.0
FW-G2	Outside	Intranet-Subnet	10.53.2.66	255.255.255.192
FW-G1	Inside	Intranet-Subnet	10.53.2.65	255.255.255.192
FW-G1	Outside	G1-Subnet	10.53.2.138	255.255.255.248
SW-R1	VLAN1	R1-Subnet	10.53.2.131	255.255.255.248
SW-R2	VLAN1	DMZ-Subnet	10.53.2.3	255.255.255.192
SW-Int	VLAN1	Internal-Subnet	10.53.1.3	255.255.255.0
SW-G2	VLAN1	Intranet-Subnet	10.53.2.67	255.255.255.192
SW-G1	VLAN1	G1-Subnet	10.53.2.139	255.255.255.248

Server Placement and Addressing

Consideration to each servers' function and desired level of accessibility was given when faced with the question of where exactly they should be placed on the JS network. Under the IEC 62443 standard, zones group physical or logical assets with similar security requirements, while Conduits restrict and control the data flows between these zones. This framework creates a defense-in-depth strategy by preventing a breach in one area from compromising the entire system.(2) (3) The JS-Sales-Server is an E-commerce server for company sales where internet users connect via HTTPS to place orders. The requirement for public internet accessibility is why this server is connected to the DMZ-Subnet. The DMZ-Subnet's placement on the red rail allows the JS-Sales-Server to be reached by the public while remaining segregated from internal resources. The RD-Gateway is a Microsoft Remote Desktop Gateway that allows authorized remote JS employees to connect to their workstations from the public Internet. The RD-Gateway was connected to the DMZ-Subnet due to its operations being public Internet facing and the defense-in-depth strategy to segregate the device from internal services. The DB-Server stores completed snack orders for processing and shipping. Connecting this server to the Intranet-Subnet as opposed to the DMZ-Subnet allowed for the safety of sensitive company and customer data. The Intranet subnet is not publicly accessible and therefore ideal for internal services. Internal Human Services information is stored on the HR-Server. Employees manage their employment data and benefits on this server. The fact that the HR-Server should not be accessed by anyone from the public internet and contains data that should remain highly secured is why it was connected to the Intranet-subnet. The AD-Server is a Microsoft Active Directory server providing authentication, authorization and accounting services to JS employees. This data need to remain secured and the server has no need to interact with traffic originating from the public internet. A connection to the Intranet-Subnet is ideal for the AD-Server. The WSA-Gateway is a Web Security Appliance providing a secure web application proxy for JS employees connecting to the public internet. While the WSA-Gateway does interact with the public internet, it is still connected to the Intranet-Subnet. The Intranet-Subnet's placement on the green rail allows the WSA-Gateway to initiate outgoing connection requests to web applications.

JS Network Server Placement and IPs				
Server Name	Subnet Name	Server IP	Subnet Mask	Default Gateway IP
JS-Sales-Server	DMZ	10.53.2.10	255.255.255.192	10.53.2.2
DB-Server	intranet	10.53.2.70	255.255.255.192	10.53.2.66
HR-Server	intranet	10.53.2.71	255.255.255.192	10.53.2.66
RD-Gateway	DMZ-Subnet	10.53.2.11	255.255.255.192	10.53.2.2
AD-Server	intranet	10.53.2.72	255.255.255.192	10.53.2.66
WSA-Gateway	intranet	10.53.2.73	255.255.255.192	10.53.2.66

Network Address Translation

NAT or network address translation is an essential step in the JS network's IP planning. It facilitates the JS network's ability to utilize private IP addresses for hundreds of devices while not confronting the issue of public IP address scarcity. The JS network only needs two public IP addresses thanks to NAT. The JS network only has three servers that require communication with the public Internet and of which, all private IP addresses must remain private to avoid security risks. JA-Sales-Server, and RD-Gateway are located on the red rail and require incoming public internet traffic to function as intended. The WSA-Gateway on the green rail sends outgoing traffic from the network to the public internet. The outside interface on Rtr-Red possesses the public IP 182.29.9.42/30. This is the IP address in which customers on the public internet can access the JS network and communicate with JS-Sales-Server to make purchases. Employees working remotely will also use this IP address to connect to RD-Gateway and access their workstations once a connection is established and employee credentials are provided. Public IP address 77.5.19.6 is provided to Rtr-Green. It is this public IP that facilitates internal employee internet accessibility via the WSA-Gateway. The choice to apply NAT on the routers and not the firewalls is supported by NIST SP 800-41, Revision 1, Guidelines on Firewalls and Firewall Policy. "Despite the popular misconception, NAT is not part of the security functionality of a firewall. The security benefit of NAT—preventing a host outside the firewall from initiating contact with a host behind NAT—can just as easily be achieved by a stateful firewall with less disruption to protocols that do not work as well behind NAT." (1) Due to the firewalls' least privilege ACLs, as well as the architecture of the red rail green rail topology segregating outbound and inbound connection requests, any NAT configuration on firewalls would serve no practical function. The security implications of configuring NAT on the firewalls for this network would be redundant at best. NAT is configured on Rtr-Red translating the destination IP address for incoming packets on the outside interface from 182.29.9.42 to 10.53.2.10:443 facilitating public internet access to the JS-Sales-Server, and from 182.29.9.42:8080 to 10.53.2.11:443 facilitating public internet access to RD-Gateway. When return traffic is received by Rtr-Red via the inside interface, source IP 10.53.2.10 is translated to 182.29.9.42, and source IP 10.53.2.11 is translated to 182.29.9.42.

To demonstrate how NAT works, I will provide a packet analysis of a few scenarios. In the first scenario, an Internet user at public IP address 4.5.6.7 connects to JS-Sales-Server to order some nuts. Packet sent by client: Source IP is 4.5.6.7, Destination IP is 182.29.9.42 port 443. The packet received by JS-Sales-Server: Source IP is 4.5.6.7, Destination IP is 10.53.2.10 port 443. In the second scenario, WSA-Gateway sets up a new HTTP connection to a public web server at IP address 22.3.4.5. Packet sent by WSA-Gateway: Source IP is 10.53.2.73 port 80, Destination IP is 22.3.4.5 port 80. The packet received by the public web Server: Source IP is 77.5.19.6 port 80, Destination IP is 22.3.4.5 port 80. In the third scenario, an Internal Client connects to HR-Server to check on their vacation time. The packet sent by internal client: Source IP is 10.53.1.10 port 443, Destination IP is 10.53.2.71 port 443. There is no NAT observed in the packet received by the HR-Server: Source IP is 10.53.1.10 port 443, Destination IP is 10.53.2.71 port 443.

Static Routes

The static routes for the JS network have been configured to promote both accessibility and security. Static routes are directions a firewall or router uses to forward packets from one subnet to another. The destination IP of a packet is examined by the router, the router searches for the ip in its routing table and receives instructions for the next hop IP address to forward the packet to when the router is not directly connected to any ip address in the subnet of the packet's destination IP. To maintain a security in depth design, not every router or firewall has a static route for every destination in the network. There are some subnets that should not be allowed to communicate with others. Communications that are necessary, allowing traffic to flow from one security zone or subnet to another, have to flow through firewalls security approval.

The Rtr-Red static route configuration consists of a pathway to the public internet where it sends default traffic to the web via next hop IP 182.29.9.41 on its outside interface. It also has a static route for traffic intended for the DMZ-Subnet via the next hop IP 10.53.2.130 on outgoing interface inside. IP address 10.53.2.130 is the FW-R1 outside interface. Rtr-Red has no other static routes because its intended purpose is simply routing inbound traffic originating from the public internet. A route to any other subnet on the JS network would result in a security risk. FW-R1 has a default static route for traffic intended for the public internet. This route sends traffic from its outside interface to next hop IP 10.53.2.129, the ip address of Rtr-Red's inside interface. FW-R1 serving as the default gateway for JS-Sales-Server and RD-Gateway required static routes to be configured to enable internal server accessibility. FW-R1 is configured with static routes to the Internal-Subnet and Intranet-Subnet via next hop ip 10.53.2.2, FW-R2's outside interface. FW-R2 includes a default static route with a next hop IP of 10.53.2.1 via its outside interface. In order to facilitate communication from the DMZ-Subnet to the Intranet-Subnet so that JS-Sales-Server has access to the DB-server, a static route to the intranet-subnet via next hop ip 10.53.1.2 was configured on FW-R2. The static route configuration for FW-G2 includes a route to the DMZ-Subnet via next hop IP 10.53.1.1, an IP belonging to FW-R2. This route enables the permitted communication between servers on the DMZ and servers on the intranet. The default gateway static route is reached via next hop ip belonging to FW-G1. There are no static routes to the G1-subnet or R-1 subnet because FW-G2 has no need to receive or send traffic to the public facing internet. FW-G1 is configured with a static route that forwards traffic to the internet via next hop IP 10.53.2.137 on Rtr-Green. This static route enables communication between the public facing internet and the WSA-Gateway. No other static route was configured on FW-G1 to maintain network segmentation and prevent potential threats originating from the internet to spread across the network if FW-G1 gets compromised. Rtr-Green is configured with a static route to facilitate communication between the G1-Subnet and intranet subnet via next hop ip of FW-G1. No other static route is configured on Rtr-Green to ensure that all replies generated from the public internet are sent to the WSA-Gateway. This ensures internal employee internet use is managed by the WSA-Gateway configured with the necessary security controls.

JS Network - Static Route Specifications						
Device Name	Destination Network	Destination Mask	Next Hop IP	Outgoing Interface	Comment	
Rtr-Red	0.0.0.0.0	0.0.0.0	182.29.9.41	G0/0	default to internet	
	10.53.2.0	255.255.255.192	10.53.2.130	G0/1	route to DMZ	
Rtr-Green	0.0.0.0.0	0.0.0.0	77.5.19.5	G0/0	default to internet	
	10.53.2.64	255.255.255.192	10.53.2.138	G0/1	route to intranet	
FW-R1	0.0.0.0.0	0.0.0.0	10.53.2.129	outside	default to Rtr-red	
	10.53.2.64	255.255.255.192	10.53.2.2	inside	to intranet	
FW-R2	0.0.0.0.0	0.0.0.0	10.53.2.1	outside	default to FW-R1	
	10.53.2.64	255.255.255.192	10.53.1.2	inside	rout to intranet	
FW-G2	0.0.0.0.0	0.0.0.0	10.53.2.65	outside	default to FW-R2	
	10.53.2.0	255.255.255.192	10.53.1.1	inside	Return traffic route to DMZ	
FW-G1	0.0.0.0.0	0.0.0.0	10.53.2.137	G0/1	Default to Rtr-green	
	10.53.1.0	255.255.255.0	10.53.2.66	G0/0	route to internal	

Firewall Rules

The firewall rules have been configured with Access Control Lists designed to regulate the flow of traffic and maintain network segmentation. The separations of subnets behind firewalls provide a safety measure that drastically decreases a threat actor's ability to compromise systems placed deeper in the network. Simply having a firewall does not provide the security necessary to prevent cyber attacks. While firewalls are important in maintaining a secure network infrastructure, misconfigurations can disrupt network traffic flow, leave your network vulnerable, and even disrupt business. Careful consideration is taken when implementing the firewall rules for the JS Network. ACLs configured on the JS firewalls were implemented with the needs of the company considered as well as how traffic should flow in the dual-path dual-layer architecture.

The position of the firewall on the network contributes greatly on what traffic is denied or permitted as well as the type of traffic. This dual-path dual-layer architecture only allows incoming TCP requests originating from the public internet on the red-rail. The green-rail only allows outgoing TCP requests to the public internet. In order to enforce the single direction traffic flows of these rails, some firewalls were configured with ACLs that deny any traffic in a direction that contradicts the dual-path dual-layer architecture. Examining the firewall rules for FW-R1, we observe a single ACL that simply denies any traffic, deny ip any any, on the inside interface. This ACL ensures that no traffic generated in the DMZ will attempt to access the public internet via route of DMZ to FW-R1 to Red-Rtr. The one way permit firewall configuration, where new connections are only allowed to originate from one side and traffic initiated from the opposite side is denied, simplifies firewall management and improves security. With firewalls configured in this manner, traffic flows in a predictable direction. New connections enter on the red rail, new connections originating from the JS network exit via the green-rail. This network design provides fewer opportunities for attackers to initiate unauthorized traffic resulting in a reduced attack surface. Another advantage to this architecture is the simpler ACL design.

With each firewall only needing to evaluate traffic in one direction, we can easily configure ACLs that meet the needs of the JS network. Firewalls examine packet information for information like source ip, destination IP, the protocol, and port numbers. This information is compared to the ACL to determine if the packet is denied or permitted. Internet users can connect to JS-Sales-Server using HTTPS over TCP with an ACL configured on a firewall's outside interface that permits traffic from any public IP address looking to form a TCP connection to the JS-Sales-Server's ip address via port 443. Internet users can connect to RD-Gateway using default TCP port for Remote Desktop Protocol (RDP) with an ACL configured on a firewall's outside interface that permits traffic from any public IP address looking to form a TCP connection to the RD-Gateway's ip address via port 3389. JS-Sales-Server can connect to DB-Server using the default PostgreSQL TCP port to submit web sales orders with an ACL configured on a firewall that permits traffic from the JS-Sales-server's IP address looking to form a TCP connection to the DB-Server's ip address via port 5432. JS Internal Clients can connect to DB-Server using the default PostgreSQL TCP port to enter sales orders received over the telephone into the database with an ACL

configured on a firewall that permits traffic from an IP address in the JS-Internal client subnet looking to form a TCP connection to the DB-Server's ip address via port 5432. JS Internal Clients can connect to AD-Server using default TCP port for Lightweight Directory Access Protocol (LDAP) to authenticate user logins with an ACL configured on a firewall that permits traffic from an IP address in the JS-Internal client subnet looking to form a TCP connection to the AD-Server's ip address via port 389. RD-Gateway can connect to AD-Server using default TCP port for Lightweight Directory Access Protocol (LDAP) to authenticate user logins with an ACL configured on a firewall that permits traffic from RD-Gateway's IP address looking to form a TCP connection to the AD-Server's ip address via port 389. RD-Gateway can connect to JS Internal Client Workstations using RDP over TCP with an ACL configured on a firewall that permits traffic from RD-Gateway's IP address looking to form a TCP connection to an IP address in the JS-Internal client subnet via port 3389. JS Internal Client Workstations can connect to HR-Server using HTTPS over TCP with an ACL configured on a firewall that permits traffic from an IP address in the JS-Internal client subnet looking to form a TCP connection to the HR-Server's ip address via port 443. JS Employees can connect to WSA-Gateway using HTTP or HTTPS over TCP to request an outgoing connection to a web server on the public Internet with an ACL configured on a firewall that permits traffic from an IP address in the JS-Internal client subnet looking to form a TCP connection to the WSA-Gateway's ip address via port 443 or 80. WSA- Gateway can connect to any public Internet web server using HTTP or HTTPS over TCP with an ACL configured on a firewall that permits traffic from WSA-Gateway's IP address looking to form a TCP connection to any IP address via port 80 or 443.

JS Network Firewall Ruleset Specifications										
(Ruleset Name = "<device>-<interface>" for Incoming packets)										
RuleSet Name	Rule #	P/D	Protocol	Source IP (First)	Source IP (Last or Mask)	Source Port	Dest IP (First)	Dest IP (Last or Mask)	Dest Port	Comments
FW-R1-Inside	1	D	IP	any	any	any	any	any	any	Deny all
FW-R1-Outside	1	P	TCP	any	any	any	10.53.2.10	255.255.255.255	443	internet traffic to JS-Sales
	2	P	TCP	any	any	any	10.53.2.11	255.255.255.255	3389	Internet traffic to RD-Gateway
	3	D	IP	any	any	any	any	any	any	Deny all
FW-R2-INSIDE	1	D	ip	any	any	any	any	any	any	Only incoming connections allowed on Red Rail
FW-R2-Outside	1	P	TCP	10.53.2.10	255.255.255.255	any	10.53.2.70	255.255.255.255	5432	JS-Sales to DB-Server via PostgreSQL
	2	P	TCP	10.53.2.11	255.255.255.255	any	10.53.2.72	255.255.255.255	389	RD-Gateway to AD-Server LDAP
	3	P	TCP	10.53.2.11	255.255.255.255	any	10.53.1.0	255.255.255.0	3389	RD-Gateway to Internal Users via RDP
	4	D	ip	any	any	any	any	any	any	
FW-G2-Inside	1	P	TCP	10.53.1.0	255.255.255.0	any	10.53.2.70	255.255.255.255	5432	Internal Users to DB-Servers
	2	P	TCP	10.53.1.0	255.255.255.0	any	10.53.2.72	255.255.255.255	389	Internal Users to AD-Servers
	3	P	TCP	10.53.1.0	255.255.255.0	any	10.53.2.71	255.255.255.255	443	Internal Users to HR-Server
	4	P	TCP	10.53.1.0	255.255.255.0	any	10.53.2.73	255.255.255.255	80	Internal Users to WSA-Gateway
	5	P	TCP	10.53.1.0	255.255.255.0	any	10.53.2.73	255.255.255.255	443	Internal Users to WSA-Gateway
	6	P	TCP	10.53.2.11	255.255.255.255	any	10.53.2.72	255.255.255.	389	RD-Gateway to AD-Server
	7	P	TCP	10.53.2.10	255.255.255.255	any	10.53.2.70	255.255.255.255	5432	JS-Sales server to DB-Server
	8	D	IP	any	any	any	any	any	any	
FW-G2-Outside	1	D	ip	any	any	any	any	any	any	Only outgoing connections allowed on Green Rail
FW-G1-Inside	1	P	TCP	10.53.2.73	255.255.255.255	any	any	any	80	WSA-Gateway to Internet HTTP
	2	P	TCP	10.53.2.73	255.255.255.255	any	any	any	443	WSA-Gateway to Internet HTTPS
	3	D	IP	any	any	any	any	any	any	
FW-G1-Outside	1	D	ip	any	any	any	any	any	any	Only outgoing connections allowed on Green Rail

Modification for High-Availability Service.

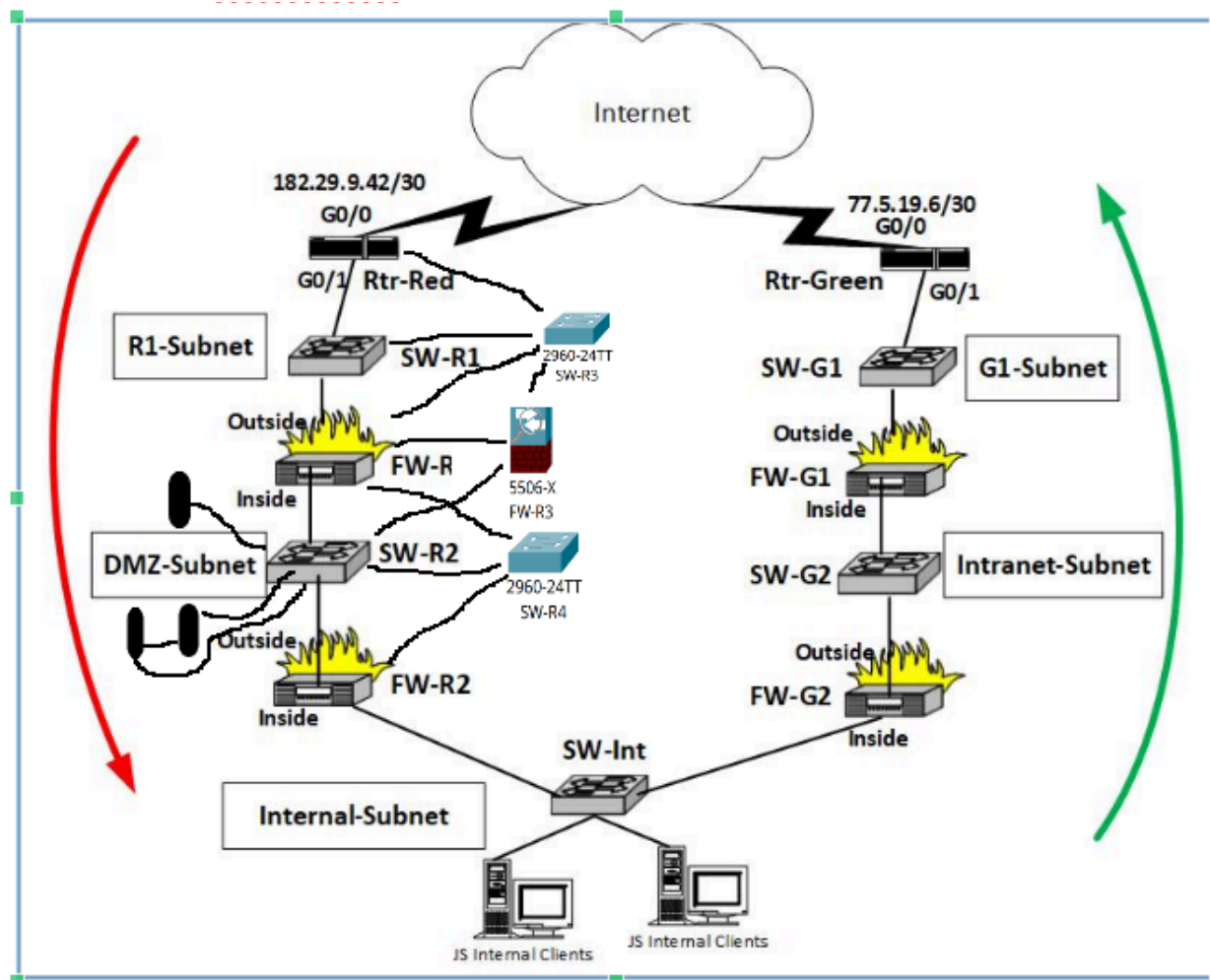
The ability to carry on with business as usual is an important consideration when the technology on the network, like all technology, is subject to required updates, maintenance, technical issues, and outright failure. To ensure that customers have constant access to the network to place orders and network issues do not result in profit loss, the JS network can be modified to avoid single point failure by adding two additional switches, one firewall and a second JS-Sales-Server or a standby server. Adding redundant switches for the DMZ subnet with redundant links between routers, switches, firewalls and servers can also provide near constant access for the customer. A virtual IP address is an IP address that is not tied to a specific physical network interface or device. It serves as a logical identifier assigned to a service, dynamically routing customers to the JS network. A Virtual IP address will allow the network to be reached by the customer via the same public IP address without a DNS change.

Switch and link redundancy will call for Spanning Tree Protocol to prevent broadcast storms, multiple frame copies, and MAC address table instability on the network. STP is a layer 2 protocol designed to prevent switching loops in networks containing redundant links. The protocol works by having multiple switches electing a root bridge, the central logical reference point of the network, and then calculating the shortest path to the root bridge. The switches then block redundant links while keeping them available as backup and automatically activating blocked links if an active link fails. (4) With switch and link redundancy added to the DMZ, STP ensures connectivity to continue without creating switching loops.

Firewall redundancy, fault tolerance, and increased availability for the JS network can be achieved with firewall clusters. Firewall clustering ensures if one firewall fails, another firewall will take over immediately with no manual intervention required. This modification would be especially impactful for the JS-Sales-Server, RD-Gateway, and Employee internet access by providing a significant reduction in downtime. Implementing firewall clusters for the JS network can be accomplished with ASA supported failover modes Active/Active mode or Active/Passive mode. In Active/Active mode, firewall A and B are both actively forwarding traffic at the same time. Some advantages to Active/Active mode include better hardware utilization with both firewalls being actively used, traffic can be divided between devices resulting in better load distribution, and the increased throughput due to the combination of processing power available. While this does sound appealing, Active/Active mode comes at the cost of complicated configurations, complex troubleshooting, and higher administrative overhead. (5) In Active/Passive mode, only one device processes traffic. The passive firewall is on standby continuously receiving configuration updates, state information, and session information. "When a failover occurs, the Active unit fails over to the Standby unit, which then becomes Active." (6) The beauty of Active/Passive cluster really is in its simplicity, easy troubleshooting, lower chance of synchronization issues. The biggest disadvantages to an Active/Passive firewall cluster modification would be the hardware underutilization, and no load sharing.

The JS network should deploy Active/Passive firewall clusters rather than Active/Active clusters. Active/Passive clustering provides high availability, simplified

management, predictable failover behavior, and lower operational risk while still ensuring that Internet users can continue accessing the JS-Sales-Server during firewall failures. Although Active/Active clustering offers better hardware utilization and increased throughput, the additional complexity is not justified by the size and performance requirements of the JS Network. Therefore, Active/Passive clustering represents the most cost-effective and operationally efficient solution for achieving firewall redundancy and maintaining business continuity.



The JS Network Diagram

Conclusion

Jasper, the Dual-Path Dual-Layer network architecture proposed in this report offers Joe's Snacks a uniquely secure, highly scalable, and exceptionally efficient foundation for its expanding e-commerce operations. By strategically separating traffic into the incoming "Red Rail" and outgoing "Green Rail" pathways, this design inherently neutralizes complex lateral threats before they can reach your critical assets. The intentional allocation of five distinct subnets guarantees that public-facing resources like the JS-Sales-Server and RD-Gateway remain properly isolated in the DMZ-Subnet, while sensitive databases and internal client workstations are safely tucked away inside the Intranet and Internal subnets. Coupled with precise Network Address Translation (NAT) configurations and strict, one-way firewall rulesets, this topology seamlessly supports remote employees and customer orders without exposing your corporate core to web application or credential-stuffed attacks. Furthermore, the proposed high-availability modifications ensure that your digital storefront remains resilient against single points of failure, keeping business fluid and uninterrupted.

A sophisticated architecture requires an analyst who understands how these configurations translate into real-world operational security. My technical foundation in advanced digital forensics and network administration allows me to look at your infrastructure through the eyes of an attacker, ensuring that every static route, ACL line, and subnet boundary is defensively sound. Navigating complex network topologies in enterprise environments has taught me to balance tight security parameters with the practical needs of everyday end users. I have dedicated myself to mastering the exact methodologies required to protect growing companies from modern threat actors. Joe's Snacks deserves a network designed not just for today's sales, but for tomorrow's security, and I am entirely equipped and ready to build that future for you.

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